

## Element H

|   | requirement                                                                     | testing setup and procedure                                                                                                                                                                                                                                                                                                             | test cases and number of trials                                                                                                      | expected range of results                                                                                                 | pass/fail criteria, goal                                                                                                                                                               |
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| 1 | We must create a detection system that successfully detects an element of fire. | <ol style="list-style-type: none"> <li>1) Set up the detection system.</li> <li>2) Light a small piece of wood on fire, then put it out, allowing the smoke to disperse.</li> <li>3) Hold the smoking piece of wood close to the detector part of the system.</li> <li>4) Review if the system goes off or not.</li> </ol>              | <ul style="list-style-type: none"> <li>- We tested at least 8 times.</li> <li>- Test until the detection system goes off.</li> </ul> | Detection system may: <ul style="list-style-type: none"> <li>- Make sound</li> <li>- Make light</li> <li>- etc</li> </ul> | <b>Pass:</b> The detection system goes off.<br><b>Fail:</b> The detection system does not work.<br><b>Goal:</b> For the detection system to successfully detect the smoke and/or fire. |
| 2 | We must create a suppression system that successfully suppresses the fire.      | <ol style="list-style-type: none"> <li>1) Set up your suppression system, making sure it's connected to the detection system.</li> <li>2) Once the detection system goes off, wait for the suppression system to go off because it is connected to the detection system.</li> <li>3) Review whether the system works or not.</li> </ol> | <ul style="list-style-type: none"> <li>- We tested about 2-4 times.</li> <li>- Test until the system works.</li> </ul>               | Suppression system may: <ul style="list-style-type: none"> <li>- Make noise</li> <li>- Create a mess</li> </ul>           | <b>Pass:</b> The suppression system works.<br><b>Fail:</b> The system doesn't go off.<br><b>Goal:</b> For the suppression system to work and successfully put out a fire.              |
| 3 | Make the model apartment and its elements realistic overall.                    | N/A                                                                                                                                                                                                                                                                                                                                     | N/A                                                                                                                                  | N/A                                                                                                                       | <b>Pass:</b> The model apartment and everything inside it look realistic compared to a real apartment.                                                                                 |

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|   |                                                                                         |                                                                                                                                                                                                                                                                        |                                                                                                                |                                                                                                                                                     | <b>Fail:</b> The apartment doesn't look realistic.<br><b>Goal:</b> For the apartment model to look as close to realistic as possible.                                                                 |
| 4 | The systems must successfully detect and suppress the fire in 5 minutes or under.       | 1) Complete the steps in testing the system.<br>2) When the testing begins, start a timer.<br>3) When the systems successfully work and finish, end the timer.<br>4) Review what the time ended with.                                                                  | - 1-3 times<br>- Test until the systems put out the fire in under 5 minutes.                                   | Systems may: <ul style="list-style-type: none"> <li>- Not activate</li> <li>- Not suppress or detect within the time limit.</li> </ul>              | <b>Pass:</b> Fire is taken care of within 5 minutes.<br><b>Fail:</b> Time limit is exceeded.<br><b>Goal:</b> To have the fire detected and taken out under a total of 5 minutes.                      |
| 5 | The structure must meet the requirements of the measurements given to us in the manual. | 1) When building the structure, make sure to use a ruler and/or tape measure to measure out everything.<br>2) After each component is put together, check its measurements again.<br>3) Measure everything once it is together to check if anything needs to be fixed. | - We tested before and after each component was put together.<br>- Test after and before everything is set up. | Structure may: <ul style="list-style-type: none"> <li>- Shift when screws or other parts are added.</li> <li>- Fall apart</li> <li>- etc</li> </ul> | <b>Pass:</b> All measurements for the model are correct.<br><b>Fail:</b> Any measurement is incorrect.<br><b>Goal:</b> To have the entire model and systems with the correct measurements in the end. |
| 6 | The money that we spend on everything overall must                                      | N/A                                                                                                                                                                                                                                                                    | N/A                                                                                                            | N/A                                                                                                                                                 | <b>Pass:</b> We kept our expenses under 50\$<br><b>Fail:</b> If we spent over 50\$<br><b>Goal:</b> To spend                                                                                           |

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|   | be calculated at under 50\$.                                     |                                                                                                                                                                                                                                  |                                                                                           |                                                                                                                                 | some amount of money under 50\$ to contribute to the project.                                                                                                               |
| 7 | All furniture should be able to move within the apartment model. | <ol style="list-style-type: none"> <li>1) Place furniture in designated spots within the model.</li> <li>2) Pick up the model carefully.</li> <li>3) Tilt the model, allowing the furniture to allegedly move around.</li> </ol> | <ul style="list-style-type: none"> <li>- X1 times we tested</li> <li>- X1 test</li> </ul> | Furniture may: <ul style="list-style-type: none"> <li>- Move out of place.</li> <li>- Crashing into other components</li> </ul> | <b>Pass:</b> The furniture moves<br><b>Fail:</b> It doesn't have the ability to move.<br><b>Goal:</b> To have the furniture be able to be moved within the model apartment. |